

Integer factorisation and discrete logarithms

are the two main sources of challenging problems used in public-key cryptography.

We'll now consider the second of these.

First, we'll need some

Basic group theory

Defⁿ A group $(G, *)$ consists of a set G and a binary operation ("multiplication")

$$*: G \times G \longrightarrow G, (g, h) \longmapsto g * h$$

such that the following axioms are satisfied:

- There exists a distinguished identity element

$e \in G$ such that, for all $g \in G$, we have

$$e * g = g = g * e.$$

- For each $g \in G$, there exists a distinguished
inverse $\bar{g} \in G$ which satisfies

$$g * \bar{g} = e = \bar{g} * g.$$

- Associativity of $*$: for all $g, h, k \in G$, we have

$$(g * h) * k = g * (h * k).$$

If, in addition,

$$g * h = h * g$$

for all $g, h \in G$, then the group G is called
commutative (or abelian).

Remark: When the operation $*$ is clear from the context,
we often refer to G itself (rather than $(G, *)$)
as a group.

Fact: In any group, the identity element and the inverses
of elements are uniquely determined.

Examples:

- $(\mathbb{Z}_n, +_n)$ is a commutative group with identity element 0 and inverse $\bar{a} = n - a$ ($= (-a) \bmod n$).
 $\uparrow$ addition mod n

When we refer to $\mathbb{Z}_n$ as a group, we mean this group structure on the set $\mathbb{Z}_n$.

Exercise: Check all axioms.

- $(\mathbb{Z}_n, \cdot_n)$ is not a group for $n > 1$.
 $\uparrow$ multiplication mod n

Indeed, suppose, for the sake of contradiction, that

$(\mathbb{Z}_n, \cdot_n)$ is a group for some $n > 1$.

Let e be the identity element. Since $(\mathbb{Z}_n, \cdot_n)$ is a group,

$0 \in \mathbb{Z}_n$ admits an inverse $\bar{0}$: $e = 0 \cdot_n \bar{0}$.

$$\begin{aligned} \text{Hence: } 1 &= 1 \cdot_n e = 1 \cdot_n (0 \cdot_n \bar{0}) = (1 \cdot_n 0) \cdot_n \bar{0} \\ &= 0 \cdot_n \bar{0} \stackrel{!}{=} 0. \end{aligned}$$

This is a contradiction since $0 \not\equiv 1 \pmod{n}$

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for $n > 1$.

• Recall: $\mathbb{Z}_n^\times = \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}$

Then $(\mathbb{Z}_n^\times, \cdot)$ is a group (a commutative one, in fact)

— The identity element is 1.

— For $a, b \in \mathbb{Z}_n^\times$, $\gcd(ab, n) = 1$ so $a \cdot b$

really belongs to $\mathbb{Z}_n^\times$ for all $a, b \in \mathbb{Z}_n^\times$.

— The inverse of $a \in \mathbb{Z}_n^\times$ is precisely the (unique!) inverse of a modulo n in $\mathbb{Z}_n^\times$.

When we refer to $\mathbb{Z}_n^\times$ as a group, we mean the group $(\mathbb{Z}_n^\times, \cdot)$.

• Let X be a set. Define $\text{Sym}(X) = \{f: X \rightarrow X : f \text{ bijective}\}$

Then $(\text{Sym}(X), \circ)$ is a group, the symmetric group on X .

↑
compositions of functions

Fact: $(\text{Sym}(X), \circ)$ is commutative

$$\Leftrightarrow |X| \leq 2$$

- $\text{GL}_d(\mathbb{Z}_n)$ = invertible $d \times d$ matrices over $\mathbb{Z}_n$
 is a gp w.r.t. matrix multiplication over $\mathbb{Z}_n$
 $\leadsto$ Hill cipher, higher dimensional affine cipher

Powers of an element in a group

Let $(G, *)$ be a group and let $g \in G$.

We define the powers g^n ($n \in \mathbb{Z}$) as follows:

- $g^0 := e$ = identity element
- $g^n := g^{n-1} * g$ for $n \geq 1$
- $g^{-n} := \bar{g}^n$ for $n \geq 0$. (so $g^{-1} = \bar{g}$ is the inverse of g)

Fact: $g^m * g^n = g^{m+n}$ and $(g^m)^n = g^{mn}$ for all $m, n \in \mathbb{Z}$

Ex: In $\mathbb{Z}_q^\times$, $2^5 = 5 = 2^{-1}$.

Defⁿ The order of a group $(G, *)$ is $|G|$.

A group G is finite if $|G| < \infty$.

Defⁿ & Fact: Let $(G, *)$ be a finite group and let $g \in G$. Then there exists $m \geq 1$ with $g^m = e$.

The order of g (within the group) is the smallest such value of m ; it is denoted by $\text{ord}_G(g)$.

The order of g coincides with the number of elements in the finite (!) set

$$\langle g \rangle_G := \{ g^n : n \in \mathbb{Z} \}$$

$$\stackrel{!}{=} \{ g^0, g^1, \dots, g^{\text{ord}_G(g)-1} \}$$

These elements are distinct!

Ex: In $\mathbb{Z}_{25}^*$

$$6^0 = 1, \quad 6^1 = 6, \quad 6^2 = 11, \quad 6^3 = 16,$$

$$6^4 = 21, \quad 6^5 = 1 = \text{identity element of } \mathbb{Z}_{25}^*$$

Hence: $\text{ord}_{\mathbb{Z}_{25}^*}(6) = 5$ and

$$\langle 6 \rangle_{\mathbb{Z}_{25}^*} = \{1, 6, 11, 16, 21\}.$$

Defⁿ A subgroup of $(G, *)$ is a subset $H \subseteq G$ such that (1) $e \in H$, (2) $a * b \in H$ for all $a, b \in H$, and (3) $a^{-1} \in H$ for all $a \in H$.

Fact: A subgroup H of $(G, *)$ is a group itself with multiplication "inherited" from $(G, *)$.

Fact: $\langle g \rangle_G$ is a subgroup of $(G, *)$ for each $g \in G$.

Lagrange's theorem

Let $(G, *)$ be a finite group and let $H \leq G$ be a subgroup. Then $|H|$ divides $|G|$.

Cor: $\text{ord}_G(g)$ divides $|G|$ for each $g \in G$.

Pf: $\text{ord}_G(g) =$ order of the subgroup $\langle g \rangle_G$ of G ,

($\rightarrow$ Have another look at the second proof Euler's Totient Theorem from Lecture 15.)

Cyclic groups

Defⁿ A group $(G, *)$ is cyclic if there exists an element $g \in G$ with $G = \langle g \rangle_G$; such an element is called a generator of the group.

Remark: Every cyclic group is commutative but not vice versa.

Ex: $\mathbb{Z}_n$ (i.e. $(\mathbb{Z}_n, +)$) is cyclic for every n : $\mathbb{Z}_n = \langle 1 \rangle_{\mathbb{Z}_n}$

Suppose that $(G, *)$ is a cyclic group of finite order n with generator g .

Then G consists of the distinct (!) elements

$$g^0, g^1, \dots, g^{n-1}$$

and the group operation $*$ is given by

$$g^i * g^j = g^{(i+j) \bmod n}$$

the inverse of g^i is $g^{-i} = g^{(-i) \bmod n}$.

(This means that $(G, *)$ "looks like" $(\mathbb{Z}_n, +)$.

Formal approach: $\rightarrow$ Group isomorphism.)